Day 1: Tools for Troubleshooting Theory (1 hour): Overview of tools (multimeter, POST card, software diagnostics).

1.List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in cach case.

ANS:

Three Real-Life Scenarios Where a Multimeter Is Used

No.	Scenario	How the multimeter helps
1.	Computer power supply issue	Measures the output voltage of the power supply to check whether it provides the correct voltage.
2.	Smartphone charging problem	Checks voltage at the charger or charging circuit to help identify power supply problems.
3.	Laptop battery problem	Measures the battery voltage to determine whether it is supplying the expected voltage.

**2.Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.**

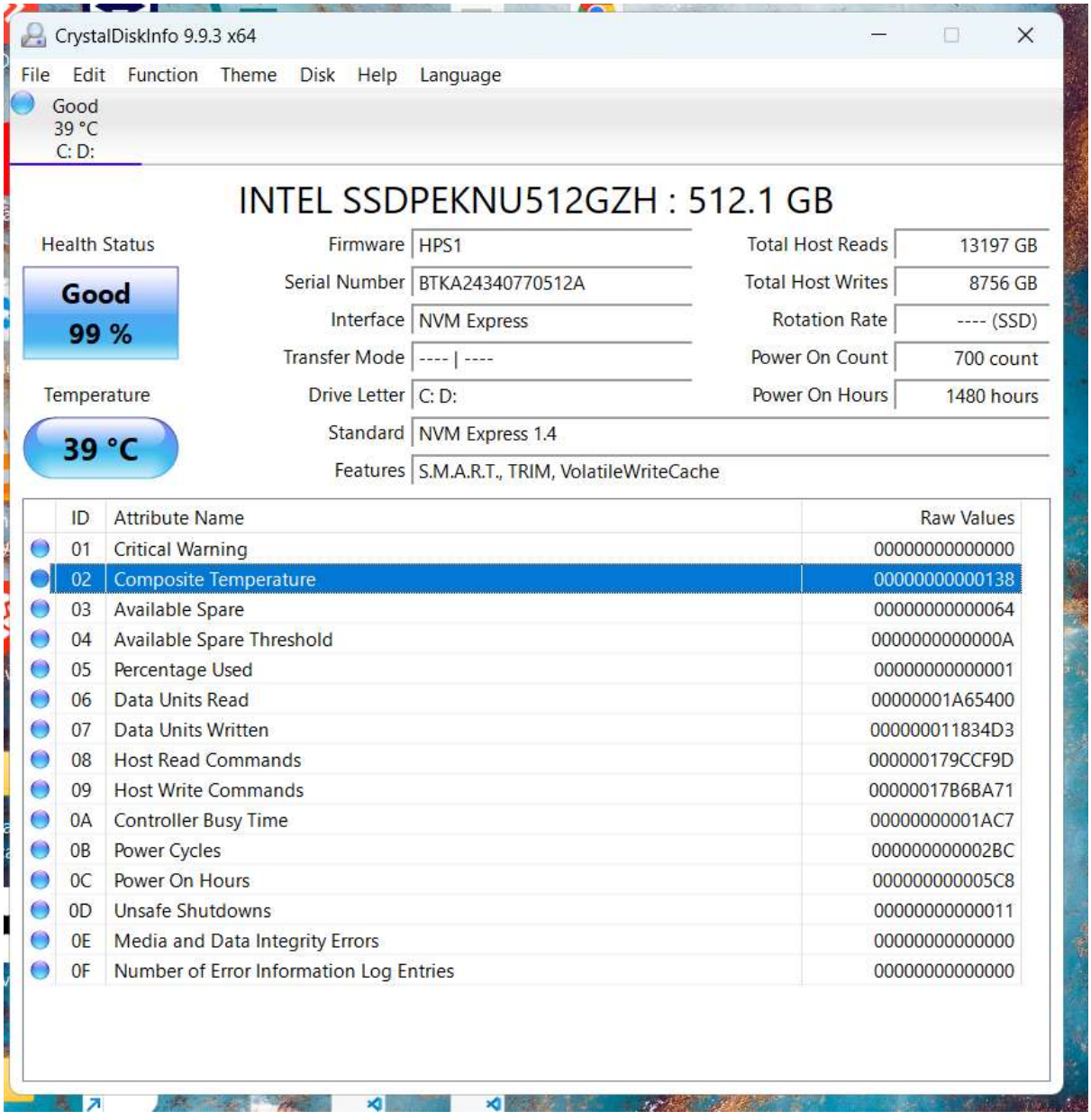
ANS:

A POST card is connected to the motherboard to diagnose hardware problems when a computer fails to start.
It displays hexadecimal POST codes on an LED screen as the computer performs its Power-On Self-Test.
The codes help identify possible hardware faults, such as RAM, CPU, or graphics-related problems.
By checking the displayed code against the motherboard manual, a technician can identify the likely cause of a boot failure.

**3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.**

ANS:



4.Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

ANS:

Comparison of Computer Diagnostic Tools

Tool	What it checks	Example Problem Detected
Multimeter	Measures voltage, current, and resistance in electrical circuits.	A faulty power supply that provides incorrect voltage to the motherboard.
POST Card	Displays diagnostic POST codes during computer startup to help identify hardware initialization problems.	A computer fails to start because of a possible RAM or CPU initialization issue.
Software Diagnostics (e.g., CrystalDiskInfo, HWMonitor)	Checks hard drive health, temperature, voltage, and other available system sensor readings.	CrystalDiskInfo detects a warning status or SMART errors indicating a possible hard drive failure.